

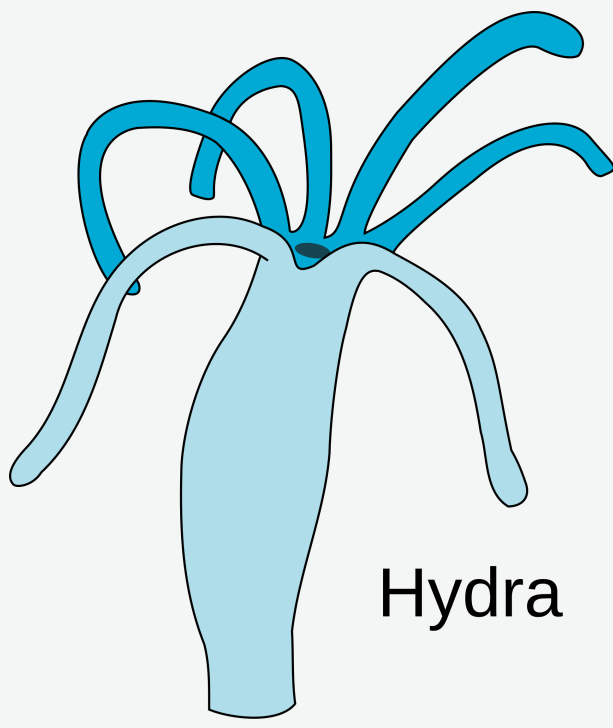
What Does Source Density Add to Models of Hydra Patterning and Regeneration?

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1 Motivation

1.1 Hydra as a patterning problem



A *Hydra* regenerates a single head after amputation and keeps its oral–aboral axis. Head formation is organised by a localised Wnt/ β -catenin signal. In activator–inhibitor models, the head is represented by a localised peak of the activator **u**, controlled by a long-range inhibitor **v**.

1.2 Head grafting

HEAD-GRAFT SCHEMATIC
head tissue transplanted into the body column
→ secondary axis, outcome depends on graft position

A grafted head can induce a second axis. Whether it does depends on *where* in the host body column it is placed.

One model must separate three phenomena: regeneration, pattern selection, and positional memory.

2 What the activator–inhibitor model already does

2.1 Production terms change the phase portrait

$$\begin{aligned}\partial_t u &= D_u \Delta u + \frac{au^2 + q_u}{v} - \mu_u u + p_u, \\ \partial_t v &= D_v \Delta v + bu^2 - \mu_v v + p_v.\end{aligned}$$

The three production terms are not interchangeable:

- p_v : basal inhibitor production; regularises the resting state,
- p_u : external activator input,
- q_u : activator input inside inhibitor regulation.

Homogeneous equilibria satisfy the cubic

$$P(u) = \mu_v bu^3 - (a\mu_v + bp_u)u^2 + \mu_u p_v u - (\mu_v q_u + p_u p_v) = 0.$$

PHASE-PORTRAIT PLACEHOLDER
one vs. three homogeneous states
nullclines + stability markers

1 state high activator production can remove the threshold.
3 states lower rest + unstable threshold + upper active state.
DII an outer kinetically stable state can become diffusion unstable.

2.2 Inhibitor production creates a threshold

From here on $p_u = q_u = 0$:

$$\begin{aligned}\partial_t u &= D_u \Delta u + \frac{au^2}{v} - \mu_u u, \\ \partial_t v &= D_v \Delta v + bu^2 - \mu_v v + p_v.\end{aligned}$$

E_- is the activation threshold; E_+ is the candidate pattern-forming state.

Rest state $E_0 = (0, p_v/\mu_v)$. Two positive states $E_{\pm}$ exist when

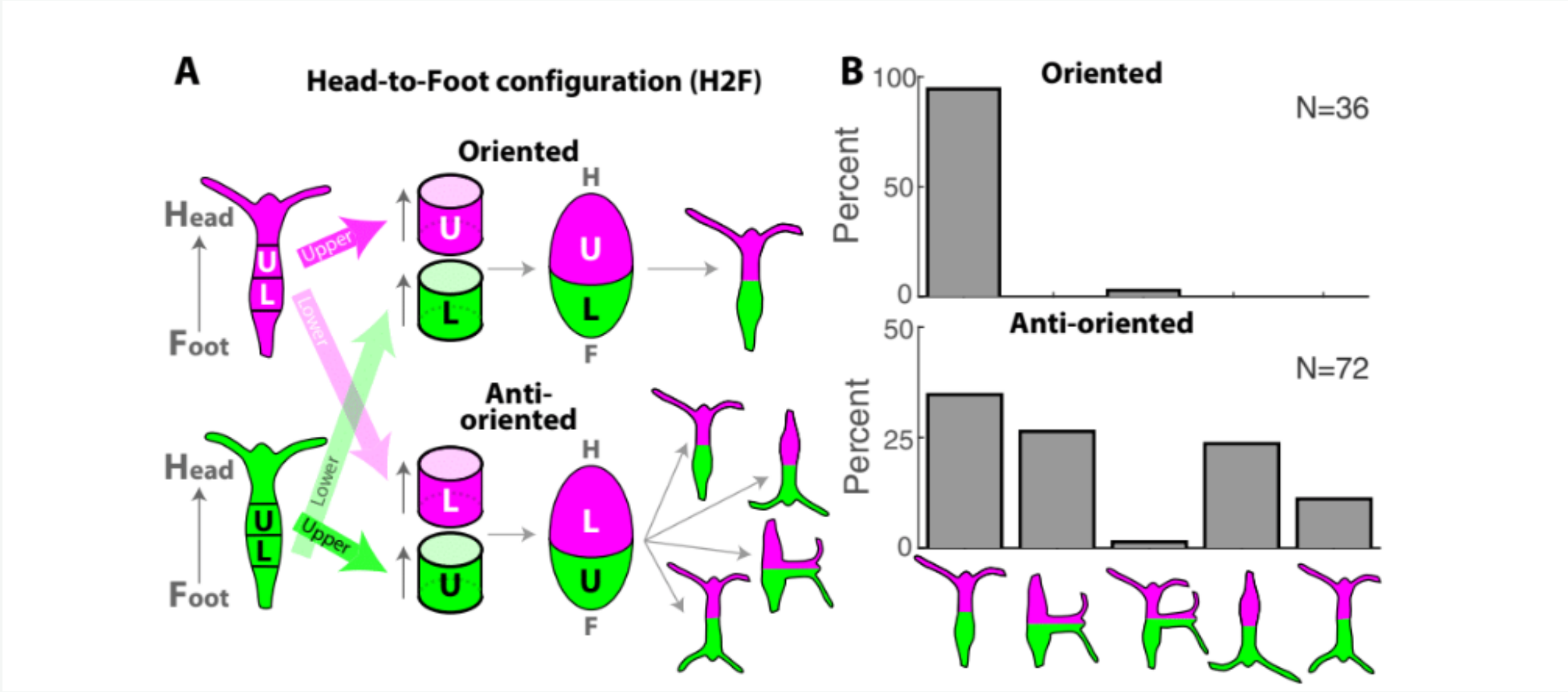
$$\begin{aligned}\Delta_h &= a^2 \mu_v^2 - 4 \mu_u^2 b p_v > 0, \\ u_{\pm} &= \frac{a \mu_v \pm \sqrt{\Delta_h}}{2 \mu_u b}, \quad v_{\pm} = \frac{a}{\mu_u} u_{\pm}.\end{aligned}$$

Basal inhibitor production already gives a stable rest state, bistability, and a finite-amplitude trigger.

Source density is not needed to make a head — but the model, not the wound, decides where the head ends up.

3 Positional memory and source density

3.1 Grafting exposes missing information



Head-to-foot grafts, oriented vs. anti-oriented. Livshits et al. (2022). K. Keren group. Reversing the orientation of comparable tissue pieces changes the distribution of outcomes. Near the resting state, **u** and **v** alone do not distinguish an upper-body fragment from a lower-body one.

Which state variable records the original position and orientation of the grafted tissue?

3.2 Prescribed $\rho(x)$ as tissue competence

PROFILE PLACEHOLDER
oriented and anti-oriented piecewise $\rho(x)$

A fixed source-density profile encodes tissue origin before the activator is present and biases where a wound-triggered peak forms (source-gradient idea, Suzuki & Takagi 2011).

Source density can hold positional memory for a long time — without making the graft-selected peak truly stationary.

Take-home message

- p_v is enough for a rest state, an activation threshold, and patterning.
- The model selects head positions; wounds elsewhere give transient peaks.
- ρ adds inherited positional information for grafted tissue.
- Dynamic ρ gives metastability: long-lived anchoring, then slow reorganisation.

2.3 Regeneration needs a finite wound

JULIA SIMULATION PLACEHOLDER
weak versus strong localised perturbation

A weak perturbation returns to rest. A sufficiently strong localised perturbation crosses E_- and produces a persistent activator peak.

2.4 Grafts: the model chooses the position

JULIA SIMULATION PLACEHOLDER
wound away from preferred position
transient peak at the wound → drift to boundary / middle / one third

A second peak can be initiated by a wound, but its final position is not arbitrary: a peak created away from a preferred location drifts to a position selected by domain size and unstable modes.

2.5 Pattern selection depends on history

JULIA SIMULATION PLACEHOLDER
same final domain, two preparation histories
direct DII: many peaks | growth continuation: one peak

Starting near the diffusion-unstable state on a large domain selects several peaks. Growing the domain from a small one preserves a single established peak in the same final parameter regime.

3.3 Dynamic source-density model

$$\begin{aligned}\partial_t u &= D_u \Delta u + \rho \frac{au^2}{v} - \mu_u u, \\ \partial_t v &= D_v \Delta v + b\rho u^2 - \mu_v v + p_v, \\ \tau \partial_t \rho &= D_\rho \Delta \rho + u - \rho.\end{aligned}$$

- $\rho(0, x)$ stores the grafted configuration,
- τ sets the lifetime of memory,
- D_ρ communicates positional information in space.

3.4 Dynamic ρ gives metastable memory

JULIA SIMULATION PLACEHOLDER
long-lived peak near the graft + peak position $x_{\max}(t)$

The inherited ρ profile anchors a peak near the graft on the fast u – v time scale. As ρ relaxes, the peak drifts toward a position preferred by the activator–inhibitor subsystem.

3.5 Diffusion sets the spatial reach of memory

PARAMETER-SWEEP PLACEHOLDER
memory time versus D_ρ and τ

D_ρ too small: memory stays sharply localised. Intermediate: a usable gradient forms. Too large: spatial information is erased.

Source density is not needed to make a head.

It may be needed to remember where a head should form.

Outlook. A non-diffusing head-identity variable h with hysteresis could separate transient activator peaks from stable morphological commitment.